

Hadrian Lazic

(Systems Low-Level C/C++/Rust Software Engineer)

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PROFESSIONAL SUMMARY

Performance-focused Low-Level Systems Software Engineer with extensive experience in C, C++, and Rust. Specializes in mechanical sympathy, microarchitecture-aware design, dynamic/static memory management, dynamic analysis, and low-latency instruction pipelines. Proven track record of open-source library architecture (23,000+ downloads), runtime VM design, and deep technical evaluation of low-level memory safety and correctness invariants across ISO C, modern C++, and unsafe Rust.

TECHNICAL SKILLS

- **Languages:** C (ANSI C89, C11, C23), C++ (C++17/20), Rust (Unsafe, Raw Pointers, SIMD Intrinsics), Inline Assembly (x86_64, ARM, MIPS, RISC-V), SQLite, Java, and Python.
- **Systems Architecture & Hardware Sympathy:** Memory Hierarchy Optimization (L1/L2 Cache Residency, 64-Byte Alignment, Data-Oriented Design), CPU Pipeline Tuning (Hardware Prefetching, Branchless Bit Manipulations), Instruction-Level Parallelism (ILP), Arena/Stack Allocation, Zero-Allocation Data Structures.
- **Verification & AI Evaluation:** Code Auditing, Static/Dynamic Analysis, Memory Safety Verification, Edge-Case Synthesis, Technical Documentation, ML Research Publishing.
- **Tooling & Build Systems:** GCC, Clang, CMake, xmake, Cargo, GDB/LLDB, AddressSanitizer (ASan), UndefinedBehaviorSanitizer (UBSan), Valgrind, perf, Linux (Fedora, Debian, Alpine, FreeBSD), Git, Neovim/LSP (clangd, rust-analyzer).

PROJECTS

BitmaskWindow & Zavod-Lang

C/C++ & Rust Virtual Machine Systems

- **Opcode Interpreter (Zavod-Lang):** Built a high-throughput bytecode virtual machine and register-focused execution engine tuned for L1 instruction cache locality and minimal branch mispredictions.
- **Zero-Allocation Bitmask Generator (BitmaskWindow):** Designed a data-oriented, header-level memory window isolation library leveraging compile-time bitwise operations to bypass heap allocation overhead completely.

Laststraw & Laststraw-core

Solo Author & Maintainer (17,800+ Downloads)

- **Native High-Performance GUI Framework:** Engineered a lightweight, hardware-accelerated desktop framework replacing web-runtime engines (Electron/WASM) with direct binary-level rendering pipelines.
- **API Design & Adoption:** Designed zero-cost abstraction C/Rust bindings and clear technical specifications, driving 17,800+ crate installations.

TerimalRtdm

Solo Author & Maintainer (3,000+ Downloads)

- **Low-Latency TUI Engine:** Developed an asynchronous, zero-overhead terminal interface engine operating via direct ANSI/ASCII escape sequence buffer manipulations.
- **Pipeline Saturation:** Optimized internal event loops and I/O polling mechanisms to prevent CPU instruction pipeline stalls while enforcing tight stack memory limits.

Machine Learning Research Preprint

Solo Researcher & Author

- **AI/ML Technical Publication:** Authored and published peer-reviewed independent research on *Preprints.org*, focusing on algorithmic efficiency, machine learning evaluation, and memory-constrained execution models.

EDUCATION

B.S. in Computer Science at San Diego State University (SDSU) – San Diego, CA

Incoming Fall 2026